SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 5 Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
   Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
   Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
   Estimate the probability of rolling a 6 using MLE.
5. Given:
   Input (x) = 4
   Weight (w) = 2
   Actual Output = 10
   Prediction:
   y = w × x
   1. Find the predicted output.
   2. Calculate the error.
   3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I _(K Giridhar / 192425160) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

K.Giridhar

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. Coin Toss – Maximum Likelihood Estimation (MLE)

Given:

- Outcomes: **HHTHHHTHHT**
- Total tosses (n) = 10
- Number of Heads (k) = 7

Formula

For a Bernoulli/Binomial experiment, the MLE of the probability of success is:

$$\hat{p} = \frac{k}{n}$$

where:

- k= Number of successes (Heads)
- n= Total number of trials

Calculation

$$\hat{p} = \frac{7}{10} = 0.7$$

Interpretation

The estimated probability of getting a Head is **0.7 (70%)**. This means that, based on the observed data, the coin appears to produce Heads about 70% of the time.

Answer: 0.7

2. Probability of Passing an Exam

Given:

- Total students = 50
- Students passed = 42

Formula

$$\hat{p} = \frac{\text{Passed}}{\text{Total Students}}$$

Calculation

$$\hat{p} = \frac{42}{50} = 0.84$$

Interpretation

The estimated probability that a student passes the exam is **0.84 (84%)**.

Answer: 0.84

3. Probability of a Defective Bulb

Given:

- Total bulbs = 100
- Defective bulbs = 8

Formula

$$\hat{p} = \frac{\text{Defective Bulbs}}{\text{Total Bulbs}}$$

Calculation

$$\hat{p} = \frac{8}{100} = 0.08$$

Interpretation

The estimated probability that a randomly selected bulb is defective is **0.08 (8%)**.
Answer: 0.08

4. Probability of Rolling a 6

Given:

- Total rolls = 60
- Number of sixes = 18

Formula

$$\hat{p} = \frac{\text{Number of Sixes}}{\text{Total Rolls}}$$

Calculation

$$\hat{p} = \frac{18}{60} = 0.30$$

Interpretation
The estimated probability of rolling a 6 is **0.30 (30%)**. Since this is greater than the probability for a fair die ($1/6 \approx 0.167$), the die appears to be biased toward rolling a 6.
Answer: 0.30

5. Prediction, Error, and Weight Update

Given

- Input (x) = 4
- Weight (w) = 2
- Actual Output = 10
- Learning Rate (α) = 0.01
- Gradient = 4

Step 1: Find the Predicted Output
The prediction equation is:

$$y = w \times x$$

Substitute the given values:

$$y = 2 \times 4 = 8$$

Predicted Output = 8

Step 2: Calculate the Error
Error is the difference between the actual output and the predicted output.

$$\begin{aligned} &\text{Error} = \text{Actual Output} - \text{Predicted Output} \\ &= 10 - 8 = 2 \end{aligned}$$

A positive error means the prediction is lower than the actual value.
Error = 2

Step 3: Update the Weight
In gradient descent, the weight is updated using:

$$w_{\text{new}} = w - \alpha \times \text{Gradient}$$

Substitute the values:

$$\begin{aligned} &w_{\text{new}} = 2 - (0.01 \times 4) \\ &= 2 - 0.04 \\ &= 1.96 \end{aligned}$$

Updated Weight = 1.96